

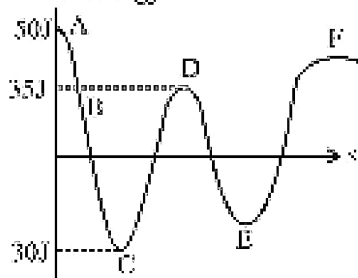
PHYSICS

1) Which of the following forces is not conservative?

- (1) $\vec{F} = 3\hat{i} + 4\hat{j}$
- (2) $\vec{F} = 3x\hat{i} + 4y\hat{j}$
- (3) $\vec{F} = 3y\hat{i} + 4x\hat{j}$
- (4) $\vec{F} = 3x^2\hat{i} + 4y^2\hat{j}$

2) An object is in the region where potential energy is changing as shown in the figure. Its total mechanical energy is 30 J. If object is initially at C, then select the correct statement.

Potential energy



- (1) Object will be able to reach at E.
- (2) At point C kinetic energy of object is 50 Joules
- (3) Object will be oscillating between B and D.
- (4) Object can reach at D.

3) Match the columns.

	Column I		Column II
(1)	Work done by a man in lifting a bucket out of well by means of a rope tied to the bucket	(P)	Work done positive
(2)	Work done by friction on a body sliding down an inclined plane	(Q)	Work done negative
(3)	Work done by the resistive force of air on an oscillating pendulum in bringing to rest	(R)	Zero work done

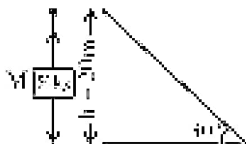
(4)	Work done by an applied force on a body moving on a rough horizontal plane with uniform velocity		
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- (1) (1) - (R), (2) - (Q), (3) - (Q), (4) - (P)
 (2) (1) - (Q), (2) - (P), (3) - (R), (4) - (P)
 (3) (1) - (P), (2) - (R), (3) - (R), (4) - (Q)
 (4) (1) - (P), (2) - (Q), (3) - (Q), (4) - (P)

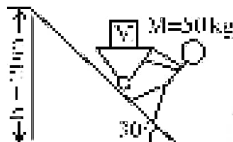
4) An object of mass 1000 g initially at rest experiences a time dependent force $\vec{F} = (2t\hat{i} + 3t^2\hat{j})\text{N}$. The power generated by the force at time t is :

- (1) $(2t^2 + 18t^3)\text{W}$
 (2) $(3t^3 + 5t^5)\text{W}$
 (3) $(2t^3 + 3t^5)\text{W}$
 (4) $(2t^2 + 3t^3)\text{W}$

5) A body of mass 50 kg is lifted to a height of 20 m from the ground in two different ways as shown in figure. The ratio of work done against the gravity in both the respective case will be :-



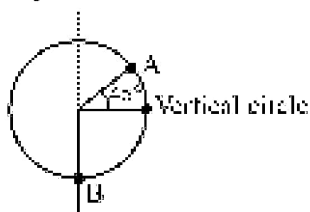
Case (1) : Pulled straight up



Case (2) : Along the ramp

- (1) 2 : 1
 (2) $\sqrt{3}$: 2
 (3) 1 : 1
 (4) 1 : 2

6) A body of mass m kg slides from rest along the curve of vertical circle from point A to B in frictionless path. The velocity of the body at B is :-

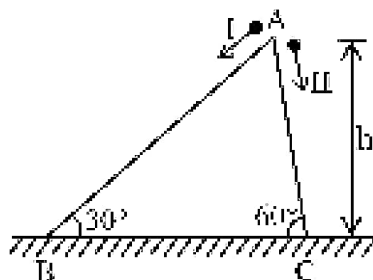


[$R = 14$ m, $g = 10$ m/s² and $\sqrt{2} = 1.4$]

- (1) 10.6 m/s

- (2) 19.8 m/s
- (3) 16.7 m/s
- (4) 21.9 m/s

7) Two inclined frictionless tracks one gradual and the other steep, meet at A from where two stones



are allowed to slide down from rest, one on each track.
of the time taken by stones to reach at the bottom.

Find the ratio

- (1) $\frac{1}{\sqrt{3}}$
- (2) $\frac{1}{2}$
- (3) $\frac{1}{\sqrt{2}}$
- (4) $\sqrt{3}$

Common Content for Question No. 8 to 9

A windmill core principle is converting the kinetic energy of wind into mechanical energy. Windmills have blades or vanes that are designed to catch the wind. As the wind blow against the blades, it creates a force that causes them to spin or rotate around a central shaft. The rotating central shaft is connected to a mechanism that converts the rotational energy into useful work.

The blades of a windmill sweep out a circle of area A and the wind (density ρ) flows at a velocity 'V' perpendicular to the circle, then :-

8) What is the mass of the air passing through it in time 't' ?

- (1) AV
- (2) ρAV
- (3) ρAVt
- (4) Information insufficient

9)

What is the kinetic energy of the air passed in time t ?

- (1) $\frac{1}{2} A^2 V^2$
- (2) $\frac{1}{2} \rho A^2 V^2 t$
- (3) $\frac{1}{2} \rho A V^3 t$
- (4) $\frac{1}{2} \rho^2 A^2 V^2$

10) Assume that the windmill converts 25% of the winds kinetic energy (in above question) into electrical energy. Calculate the electric energy produced in this process.

- (1) $\frac{1}{8}\rho A^3 V^3 t$
- (2) $\frac{1}{8}\rho^2 A^2 V^2 t$
- (3) $\frac{1}{8}\rho A V t$
- (4) $\frac{1}{8}\rho A V^3 t$

11) Calculate electric power produced in the above process.

- (1) $\frac{1}{8}\rho A^3 V^3$
- (2) $\frac{1}{8}\rho^2 A^2 V^2$
- (3) $\frac{1}{8}\rho A V$
- (4) $\frac{1}{8}\rho A V^3$

12) On doubling the wind speed, the power produced by wind mill will become :

- (1) 2 times
- (2) 4 times
- (3) 8 times
- (4) 16 times

13) The only force acting on a block along x-axis is given by

$F = -\left(\frac{4}{x^2 + 2}\right)$ Newton. When the block moves from $x = -2\text{m}$ to $x = 4\text{m}$, the change in kinetic energy of block is :-

- (1) Positive
- (2) Negative
- (3) Zero
- (4) May be positive or negative

14) A particle moving in a circle of radius R with a uniform speed takes time T to complete one revolution. If this particle were projected with the same speed at an angle θ to the horizontal, the maximum height attained by it equals $8R$. The angle of projection θ is then given by :-

- (1) $\theta = \cos^{-1} \left(\frac{gT^2}{\pi^2 R} \right)^{\frac{1}{2}}$

$$(2) \theta = \cos^{-1} \left(\frac{\pi^2 R}{2gT^2} \right)^{\frac{1}{2}}$$

$$(3) \theta = \sin^{-1} \left(\frac{4gT^2}{\pi^2 R} \right)^{\frac{1}{2}}$$

$$(4) \theta = \sin^{-1} \left(\frac{\pi^2 R}{4gT^2} \right)^{\frac{1}{2}}$$

15) A particle of mass 2 kg is moving on a circular path of radius 10 m with a speed of $\sqrt{40} \text{ ms}^{-1}$ and its speed is increasing at a rate of 3 ms^{-2} . Find the net force acting on the particle at that time.

- (1) 5 N
- (2) 10 N
- (3) 12 N
- (4) 14 N

16) If a car is moving in uniform circular motion then what should be the value of speed (v) of car, so that car cannot leave the circular path ?

(μ_s , μ_k are respectively coefficient of static friction & kinetic friction between tyres of car & road & R is radius of circular path)

- (1) $v > \sqrt{\mu_s Rg}$
- (2) $v \leq \sqrt{\mu_s Rg}$
- (3) $v = \sqrt{\mu_k Rg}$
- (4) None of these

17) **Statement-1** : When a stone of fixed mass is rotated at greater speed in a horizontal circle of fixed radius, then greater centripetal force is required.

Statement-2 : When a stone of fixed mass moving with a fixed speed is rotated in a horizontal circle of smaller radius, then greater centripetal force is required.

- (1) Both statement 1 & statement 2 are false.
- (2) Statement 1 is true but statement 2 is false
- (3) Statement 1 is false but statement 2 is true.
- (4) Both statement 1 & statement 2 are true.

18) **Assertion (A)** : When an object follows a circular path at a constant speed, then the motion of object is called uniform circular motion.

Reason (R) : The acceleration of an object in uniform circular motion is always directed away from the centre of circle.

- (1) Both (A) & (R) are correct & (R) is correct explanation of (A).
- (2) Both (A) & (R) are correct but (R) is not correct explanation of (A).
- (3) (A) is correct but (R) is incorrect.
- (4) (A) is incorrect but (R) is correct.

19) If a car is to travel with a speed v_0 along the frictionless, banked circular track of radius r_0 , the required angle of banking so that the car does not skid is

(1) $\theta = \tan^{-1} \left(\frac{v_0^2}{r_0 g} \right)$

(2) $\theta = \tan^{-1} \left(\frac{v_0}{r_0 g} \right)$

(3) $\theta = \tan^{-1} \left(\frac{r_0^2}{v_0 g} \right)$

(4) $\theta = \tan^{-1} \left(\frac{v_0^2}{2r_0 g} \right)$

20) A body is moving in a vertical circle of radius r such that string is just taut at highest point of circle, then

(1) The speed of the body at the highest point is $\sqrt{gr}$.

(2) The speed of the body at the lowest point is $\sqrt{5gr}$

(3) Both (1) & (2) are correct.

(4) Both (1) & (2) are wrong.

21) The ratio of angular velocities of minute hand and hour hand of clock is :

(1) 12 : 1

(2) 6 : 1

(3) 5 : 3

(4) 8 : 9

22) A coin placed on a rotating turntable just slips when it is placed at a distance of 9 cm from the centre. If the angular velocity of the turn table is made 4 times of original, it will just slip, if its distance from centre is :

(1) 16 cm

(2) 4 cm

(3) $\frac{1}{16}$ cm

(4) $\frac{9}{16}$ cm

23) Two concentric circular coils of ten turns each are situated in the same plane. Their radii are 20 cm and 40 cm and they carry respectively 0.2 and 0.3 ampere current in opposite direction. The magnetic field in weber/m² at the centre is

(1) $\frac{35}{4} \mu_0$

(2) $\frac{\mu_0}{80}$

(3) $\frac{7}{80}\mu_0$

(4) $\frac{5}{4}\mu_0$

24) A long solenoid is formed by winding 20 turns/cm. The current necessary to produce a magnetic field of 20 millitesla inside the solenoid will be approximately $\left(\frac{\mu_0}{4\pi} = 10^{-7} \text{ tesla} \cdot \text{metre/ampere}\right)$

(1) 8.0 A

(2) 4.0 A

(3) 2.0 A

(4) 1.0 A

25) One metre length of wire carries a constant current. The wire is bent to form a circular loop. The magnetic field at the centre of this loop is B. The same is now bent to form a circular loop of smaller radius to have four turns in the loop. The magnetic field at the centre of this new loop is

(1) 4B

(2) 16B

(3) B/2

(4) B/4

26) Two similar coils are kept mutually perpendicular such that their centres coincide. At the centre, find the ratio of the magnetic field due to one coil and the resultant magnetic field by both coils, if the same current is flown :-

(1) $1 : \sqrt{2}$

(2) $1 : 2$

(3) $2 : 1$

(4) $\sqrt{3} : 1$

27) A proton and an α -particle moving in a uniform magnetic field perpendicularly with the same speed. If proton takes 25μ sec to make 5 revolutions, then the periodic time for the α -particle would be :-

(1) $50 \mu \text{ sec}$

(2) $25 \mu \text{ sec}$

(3) $10 \mu \text{ sec}$

(4) $5 \mu \text{ sec}$

28) A deuteron of kinetic energy 50 keV is describing a circular orbit of radius 0.5 metre in a plane perpendicular to magnetic field $\vec{B}$. The kinetic energy of the proton that describes a circular orbit of radius 0.5 metre in the same plane with the same magnetic field $\vec{B}$ is

(1) 25 keV

(2) 50 keV

(3) 200 keV

(4) 100 keV

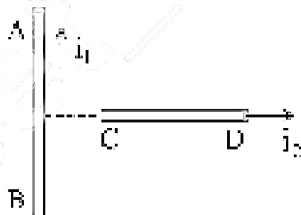
29) A proton of mass 1.67×10^{-27} kg and charge 1.6×10^{-19} C is projected with a speed of 2×10^6 m/s at an angle of 60° to the X-axis. If a uniform magnetic field of 0.104 Tesla is applied along Y-axis, the path of proton is :-

- (1) A circle of radius 0.2 m and time period $\pi \times 10^{-7}$ s
- (2) A circle of radius 0.1 m and time period $2\pi \times 10^{-7}$ s
- (3) A helix of radius 0.1 m and time period $2\pi \times 10^{-7}$ s
- (4) A helix of radius 0.2 m and time period $4\pi \times 10^{-7}$ s

30) A beam of well collimated cathode rays travelling with a speed of 5×10^6 ms⁻¹ enter a region of mutually perpendicular electric and magnetic fields and emerge undeviated from this region. If $|B| = 0.02$ T, the magnitude of the electric field is :-

- (1) 10^5 Vm⁻¹
- (2) 2.5×10^8 Vm⁻¹
- (3) 1.25×10^{10} Vm⁻¹
- (4) 2×10^3 Vm⁻¹

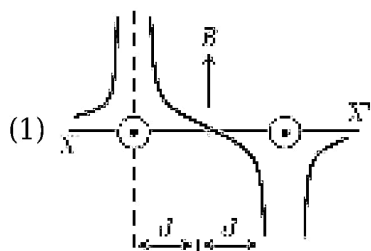
31) An infinitely long, straight conductor AB is fixed and a current is passed through it. Another movable straight wire CD of finite length and carrying current is held perpendicular to it and

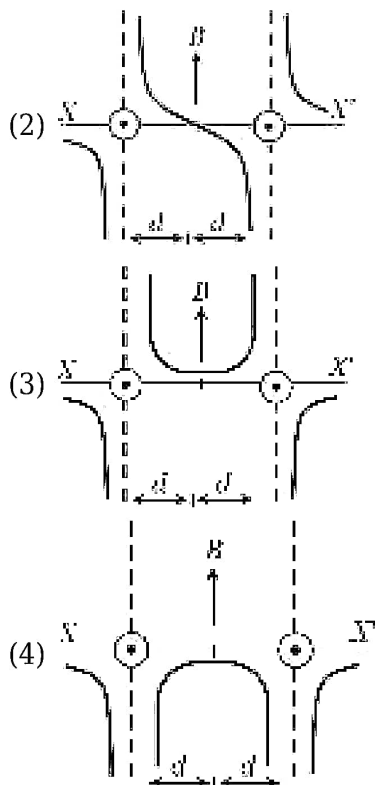


released. Neglect weight of the wire CD :-

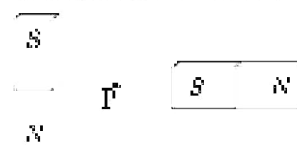
- (1) The wire CD will move upwards parallel to itself
- (2) The wire CD will move downward parallel to itself
- (3) The wire CD will move upward and turn clockwise at the same time
- (4) The wire CD will move upward and turn anti-clockwise at the same time

32) Two long parallel wires are at a distance $2d$ apart. They carry steady equal currents flowing out of the plane of the paper, as shown. The variation of the magnetic field B along the line XX' is given by :-





33) Two similar bar magnets are kept as shown in the figure. The direction of resultant magnetic



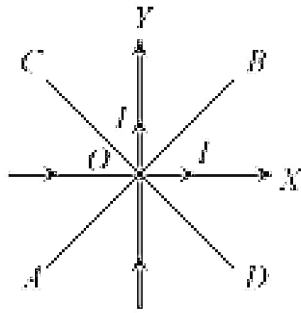
field, indicated by arrow head at the point P is (approximately)

- (1)
- (2)
- (3)
- (4)

34) Two magnets of same size and mass make respectively 10 and 15 oscillations per minute at certain place. The ratio of their magnetic moments is :-

- (1) 4 : 9
- (2) 9 : 4
- (3) 2 : 3
- (4) 3 : 2

35) Two very thin metallic wires placed along X and Y-axis carry equal currents as shown here. AB and CD are lines at 45° with the axes with origin of axes at O. The magnetic field will be zero on the



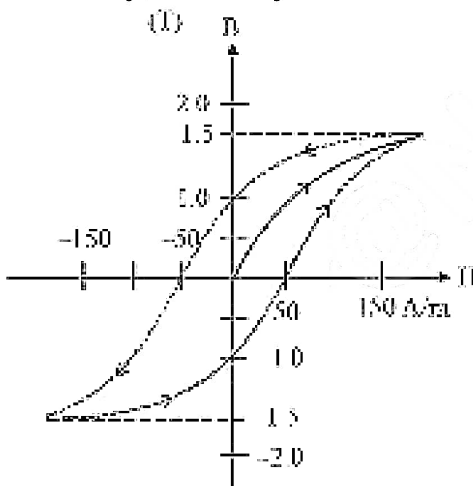
line :-

- (1) AB
- (2) CD
- (3) Only segment OB of line AB
- (4) Only segment OC of line CD

36) A bar magnet of magnetic moment 3.0 A-m^2 is placed in a uniform magnetic induction field of $2 \times 10^{-5} \text{ T}$. If each pole of the magnet experiences a force of $6 \times 10^{-4} \text{ N}$, the length of the magnet is :-

- (1) 0.5 m
- (2) 0.3 m
- (3) 0.2 m
- (4) 0.1 m

37) The figure gives experimentally measured B vs. H variation in a ferromagnetic material. The retentivity, coercivity and saturation magnetic flux density respectively, of the material are :



- (1) 150 A/m, 1.0 T and 1.5 T
- (2) 1.0 T, 50 A/m and 1.5 T
- (3) 1.5 T, 50 A/m and 1.0 T
- (4) 1.5 T, 50 A/m and 1.5 T

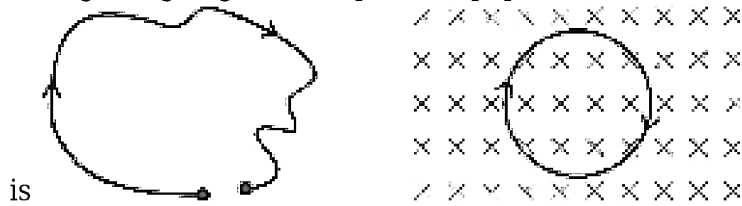
38) A magnet is suspended at an angle 60° to an external magnetic field of $5 \times 10^{-4} \text{ T}$. What is the work done by the magnetic field in bringing it in its direction ? (The magnetic dipole moment of the magnet is $20 \text{ amp} \times \text{metre}^2$) :-

- (1) $5 \times 10^{-5} \text{ J}$
- (2) $-5 \times 10^{-5} \text{ J}$

(3) $-5 \times 10^{-3} \text{ J}$

(4) $5 \times 10^{-3} \text{ J}$

39) A thin flexible wire of length L is connected to two adjacent fixed points carries a current I in the clockwise direction, as shown in the figure. When system is put in a uniform magnetic field of strength B going into the plane of paper, the wire takes the shape of a circle. The tension in the wire



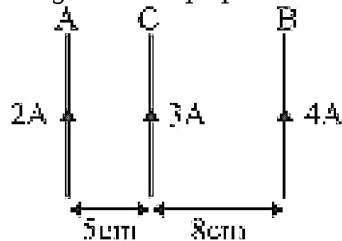
(1) IBL

(2) $\frac{IBL}{\pi}$

(3) $\frac{IBL}{2\pi}$

(4) $\frac{IBL}{4\pi}$

40) A and B are two infinitely long straight parallel conductor. C is another straight conductor of length 1m kept parallel to A and B as shown in the figure. Then the force experienced by C is :-



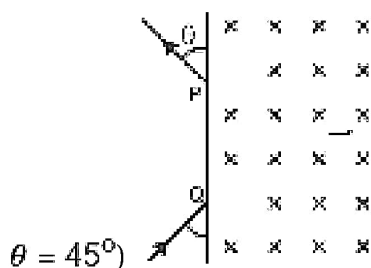
(1) towards A equal to $0.6 \times 10^{-5} \text{ N}$

(2) towards B equal to $5.4 \times 10^{-5} \text{ N}$

(3) towards A equal to $5.4 \times 10^{-5} \text{ N}$

(4) towards B equal to $0.6 \times 10^{-5} \text{ N}$

41) A particle of mass $1.6 \times 10^{-27} \text{ kg}$ and charge $1.6 \times 10^{-19} \text{ coulomb}$ enters in a uniform magnetic field of 1 Tesla as shown in the fig. The speed of the particle is 10^7 m/s . The distance PQ will be - (



(1) 0.14 m

(2) 0.28 m

(3) 0.4 m

(4) 0.5 m

42) In an experiment, a galvanometer shows full scale deflection with a series resistance of $R = 1000\ \Omega$. When a shunt resistance $S = 91\ \Omega$ is connected, the deflection reduces to half. The internal resistance of galvanometer is nearly :

- (1) $90\ \Omega$
- (2) $100\ \Omega$
- (3) $115\ \Omega$
- (4) $125\ \Omega$

43) The figure of merit (k) for galvanometer is defined as :

- (1) the current required to produce a unit deflection in the scale of the galvanometer.
- (2) the deflection produced when a unit of current passes through the galvanometer.
- (3) the deflection produced when a unit of potential difference is applied between the ends of the galvanometer.
- (4) the voltage required to produce a unit deflection in the scale of the galvanometer.

44) A cell of 1.25V and a high resistance of $1.2\ \text{k}\Omega$ are used in a circuit, to determine the resistance of a galvanometer by the half-deflection method. The galvanometer needle shows 40 divisions and its figure of merit is $25\ \mu\text{A}/\text{division}$. The value of shunt resistance, needed to cause the deflection reduced to half is :

- (1) $48\ \Omega$
- (2) $50\ \Omega$
- (3) $56\ \Omega$
- (4) $60\ \Omega$

45) A galvanometer is used in laboratory for detecting the null point in electrical experiments. If on passing the current of 3mA it produces a deflection of 24° , its figure of merit is close to : [each division on galvanometer represents deflection equal to 2°]

- (1) $2.5 \times 10^{-4}\ \text{A/div}$
- (2) $5 \times 10^{-4}\ \text{A/div}$
- (3) $2.5 \times 10^{-4}\ \text{A/degree}$
- (4) $5 \times 10^{-4}\ \text{A/degree}$

CHEMISTRY

1) The osmotic pressure of urea solution is 1.64 atm and that of sucrose solution is 2.46 atm. When equal volume of the above two solutions are mixed, the osmotic pressure of the resulting solution is :

- (1) 1.64 atm
- (2) 2.46 atm
- (3) 4.10 atm
- (4) 2.05 atm

2) Which of the following equi molal solutions will have the maximum freezing point, assuming equal ionization in each case ?

- (1) $[\text{Fe}(\text{H}_2\text{O})_6]\text{Cl}_3$
- (2) $[\text{Fe}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$
- (3) $[\text{Fe}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$
- (4) $[\text{Fe}(\text{H}_2\text{O})_3\text{Cl}_3] \cdot 3\text{H}_2\text{O}$

3)

Assertion (A) : At same temperature the osmotic pressure of 0.1 M urea solution is less than 0.1 M NaCl solution.

Reason (R) : Osmotic pressure is not a colligative property.

- (1) Both Assertion and Reason are true
- (2) Both Assertion and Reason are false
- (3) Assertion is true but Reason is false.
- (4) Assertion is false but Reason is true.

4) $E^\circ \text{Au}^+/\text{Au}$ is 1.69 V and $E^\circ \text{Au}^{+3}/\text{Au}$ is 1.40V than $E^\circ \text{Au}^+/\text{Au}^{+3}$ will be :

- (1) 0.19 V
- (2) 2.945 V
- (3) 1.255 V
- (4) -1.255 V

5) If the total vapour pressure of the liquid mixture A and B is given by the equation: $P = 180X_A + 90$ mm Hg ; then the ratio of the vapour pressure of the pure liquids A and B is given by

- (1) 3 : 2
- (2) 4 : 1
- (3) 3 : 1
- (4) 6 : 2

6) A 5% solution (by mass) of cane sugar in water has freezing point of 271 K and freezing point of pure water is 273.15 K. The freezing point of a 5% solution (by mass) of glucose in water is.

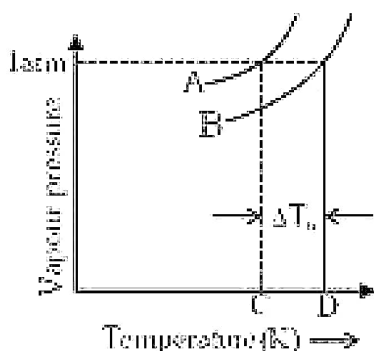
- (1) 271K
- (2) 273.15 K
- (3) 269.07 K
- (4) 277.23 K

7) **Assertion** : Isotonic solutions don't show net phenomenon of osmosis.

Reason : Isotonic solutions have equal osmotic pressure at constant temperature.

- (1) Both Assertion and Reason are true.
- (2) Assertion is false and Reason is true.

- (3) Assertion is true but the Reason is false.
 (4) Both Assertion and Reason are false.



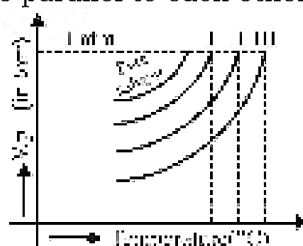
8)
 The correct match related to above graph is -

- (1) (A-solvent) (B-solution) (C- T_b), (D- T_b°)
 (2) (A-solvent) (B-solution) (C- T_b°), (D- T_b)
 (3) (A-solution) (B-solvent) (C- T_b°), (D- T_b)
 (4) (A-solution) (B-solvent) (C- T_b), (D- T_b°)

9) Henry's law constant for CO_2 in water is 1.6×10^8 Pa at 298K. The quantity of CO_2 in 500 g of water when packed under 3.2 bar pressure at 298K is.

- (1) 2.44g
 (2) 24.4g
 (3) 0.244g
 (4) 0.61g

10) The vapour pressure curves of the same non-volatile, non-electrolyte solute in the same solvent are shown. The curves are parallel to each other and do not intersect. The concentrations of



solutions are in order of :

- (1) $I < II < III$
 (2) $I = II = III$
 (3) $I > II > III$
 (4) $I > III > II$

11) Which of the following is isotonic with 15 % $\frac{W}{W}$ glucose solution (sp. gr = 1.2) :

- (1) 1 M CaCl_2
 (2) 1 M AlCl_3

- (3) 0.5 M NaCl
(4) 0.5 M urea

12) If m is the normal molecular weight and x is the degree of dissociation of $K_4[Fe(CN)_6]$, then abnormal molecular weight in the solution of salt will be :-

- (1) $\left(\frac{m}{1+2x}\right)$
(2) $\left(\frac{m}{3+x}\right)$
(3) $\left(\frac{m}{1+3x}\right)$
(4) $\left(\frac{m}{1+4x}\right)$

13) Which of the following forms ideal solution :-

- (1) $C_6H_5Cl - C_6H_5Br$
(2) $C_6H_6 - C_6H_5CH_3$
(3) Hexane-heptane
(4) All of above

14) The elevation in boiling point of a solution of 13.44 g of $CuCl_2$ (Mol. wt. 134.4) dissolved in 1 kg water is :

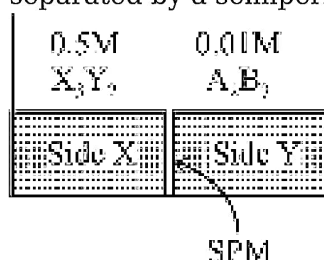
($K_b = 0.52 \text{ K molal}^{-1}$)

- (1) 0.16
(2) 0.05
(3) 0.1
(4) 0.2

15) Which one of the following aqueous solutions will exhibit highest boiling point:

- (1) 0.015 M urea
(2) 0.01 M KNO_3
(3) 0.10 M Na_2SO_4
(4) 0.015 M glucose

16) X_3Y_2 ($i = 5$) when reacted with A_2B_3 ($i = 5$) in aqueous solutions gives brown colour. These are separated by a semipermeable membrane as shown. Due to osmosis there is:



- (1) brown colour formation in side X

- (2) brown colour formation in side Y
- (3) brown colour formation in both of the sides X and Y
- (4) no brown colour formation

17) 1 kg of water under a nitrogen pressure of 1 atmosphere dissolves 0.02 g of nitrogen at 293 K. Calculate Henry's law constant :-

- (1) 7.7×10^4 atm
- (2) 7.7×10^3 atm
- (3) 2×10^{-5} atm
- (4) 2×10^{-2} atm

18)

The value of Henry's law constant for some gases at 293 K is given below. Arrange the gases in the increasing order of their solubility.

He : 144.97 kbar, H_2 : 69.16 kbar, N_2 : 76.48 kbar, O_2 : 34.86 kbar

- (1) $He < N_2 < H_2 < O_2$
- (2) $O_2 < H_2 < N_2 < He$
- (3) $H_2 < N_2 < O_2 < He$
- (4) $He < O_2 < N_2 < H_2$

19) Vapour pressure of solution containing 6 gm non volatile solute in 180 gm of water is 20 torr. If 1 mole of water is further added into the solution so that vapour pressure increases by 0.02 torr. Calculate vapour pressure of pure water.

- (1) 20.22 torr
- (2) 20.02 torr
- (3) 19.78 torr
- (4) 19.88 torr

20) When $Fe(OH)_3$ sol is prepared by condensation method then due to adsorption of which ion particles of $Fe(OH)_3$ get positive charge ?

- (1) OH^-
- (2) Fe^{+3}
- (3) H^+
- (4) Cl^-

21) Which of the following is incorrect ?

- (1) Colloidal sol of starch is an example of lyophilic sol.
- (2) Ferric hydroxide sol is prepared by direct mixing it with water.
- (3) Ferric hydroxide sol is lyophobic sol.
- (4) Colloidal sol of egg albumin is prepared by mixing egg white in to H_2O .

22) An ideal solution is prepared by mixing of 4 mol of A ($P_A^0 = 100$ mmHg) and 6 mol of B ($P_B^0 = 120$ mmHg) then mole fraction of B in vapour phase is approx.

- (1) 0.64
- (2) 0.36
- (3) 0.5
- (4) 0.25

23) For strong electrolyte, Λ_m increases slowly with dilution, it can be represented by the equation.

- (1) $\Lambda_m = \Lambda_m^0 - AC$
- (2) $\Lambda_m = \Lambda_m^0 - AC^{\frac{1}{2}}$
- (3) $\Lambda_m = \Lambda_m^0 + AC^{\frac{1}{2}}$
- (4) $\Lambda_m = \Lambda_m^0 - AC^2$

24) **Assertion** : Gum is lyophilic sol.

Reason : Because nature of attraction of gum particles with water is very high.

- (1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.
- (2) **Assertion** is true but **Reason** is false.
- (3) **Assertion** is false but **Reason** is true.
- (4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

25)

Which of the following statement is correct ?

- (1) $(\Delta_{\text{sol}} H^0)_{\text{CuSO}_4} = (\Delta_{\text{lattice}} H^0)_{\text{CuSO}_4} + (\Delta_{\text{Hydration}} H^0)_{\text{CuSO}_4}$
- (2) $(\Delta_{\text{sol}} H^0)_{\text{CuSO}_4} = (\Delta_{\text{lattice}} H^0)_{\text{CuSO}_4} - (\Delta_{\text{Hydration}} H^0)_{\text{CuSO}_4}$
- (3) $(\Delta_{\text{sol}} H^0)_{\text{CuSO}_4} = -(\Delta_{\text{lattice}} H^0)_{\text{CuSO}_4} + (\Delta_{\text{Hydration}} H^0)_{\text{CuSO}_4}$
- (4) All of the above

26) Suitable indicator for the acid-base titration of HCOOH against KOH is.

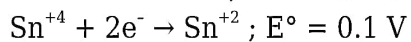
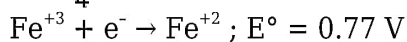
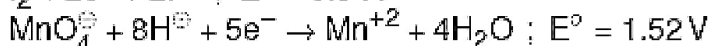
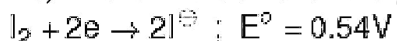
- (1) Methyl orange
- (2) Methyl red
- (3) Phenolphthalein
- (4) Thymol Blue

27) Which of the following set has maximum heat of neutralisation.

- (1) CH_3COOH and NH_4OH
- (2) HCl and NH_4OH

- (3) KOH and HCl
(4) HCN and CsOH

28) E° of some elements are given as :



Among these elements strongest reductant is :

- (1) Sn^{+2}
(2) Fe^{+2}
(3) Mn^{+2}
(4) I^-

29) Match the column I with column II for electrolysis of following solutions :

	Column-I		Column-II
(P)	CuSO_4 solution using Pt electrode	(A)	Mass of anode decreases
(Q)	CuSO_4 solution using Cu electrode	(B)	Na^+ reduce at cathode
(R)	NaCl solution using Pt electrode	(C)	$\text{O}_{2(\text{g})}$ evolved at anode
(S)	NaCl solution using Hg electrode	(D)	$\text{H}_{2(\text{g})}$ evolved at cathode

- (1) P-A, Q-C, R-D, S-B
(2) P-C, Q-A, R-D, S-B
(3) P-B, Q-D, R-A, S-C
(4) P-D, Q-B, R-C, S-A

30) If the E°_{cell} for a given reaction has a negative value, which of the following gives the correct relationships for the values of ΔG° and K_{eq} ?

- (1) $\Delta G^\circ > 0$; $K_{\text{eq}} < 1$
(2) $\Delta G^\circ > 0$; $K_{\text{eq}} > 1$
(3) $\Delta G^\circ < 0$; $K_{\text{eq}} > 1$
(4) $\Delta G^\circ < 0$; $K_{\text{eq}} < 1$

31) The cell reaction for the given cell $\text{Pt}, \text{Cl}_2 (\text{P}_1 \text{ atm}) | \text{Cl}^- | \text{Cl}_2 (\text{P}_2 \text{ atm}), \text{Pt}$ is spontaneous if -

- (1) $\text{P}_1 > \text{P}_2$
(2) $\text{P}_1 < \text{P}_2$
(3) $\text{P}_1 = \text{P}_2$
(4) $\text{P}_1 = 1 \text{ atm}$

32) The resistance of 1N solution of a salt is 50Ω . Calculate the equivalent conductance of the solution, if the two platinum electrodes in solution are 2.1 cm apart and each having an area of 4.2

cm².

- (1) 10 ohm⁻¹ cm² eq⁻¹
- (2) 0.1 ohm⁻¹ cm² eq⁻¹
- (3) 10 ohm⁻¹ m² eq⁻¹
- (4) 0.3 ohm⁻¹ m² eq⁻¹

33) At 25°C molar conductance of deci molar aqueous solution of NH₄OH is 9.54 Ω⁻¹ cm² mol⁻¹ and at infinite dilution it's molar conductance is 238 ohm⁻¹cm² mol⁻¹. The percentage of ionisation of ammonium hydroxide at the same concentration and temperature is :-

- (1) 2.08 %
- (2) 20.8 %
- (3) 4.008 %
- (4) 40.8 %

34) Match List-I with List-II.

List-I (Conversion)		List-II (Number of Faraday required)	
(A)	1 mol of H ₂ O to O ₂	(I)	3F
(B)	1 mol of MnO ₄ ⁻ to Mn ²⁺	(II)	2F
(C)	1.5 mole of Ca from molten CaCl ₂	(III)	1F
(D)	1 mol of FeO to Fe ₂ O ₃	(IV)	5F

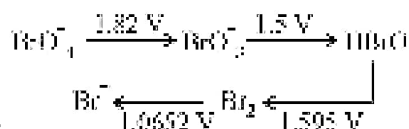
Correct answer from the given below options is :

- (1) A-II, B-IV, C-I, D-III
- (2) A-III, B-IV, C-I, D-II
- (3) A-II, B-III, C-I, D-IV
- (4) A-III, B-IV, C-II, D-I

35) $\lambda_{eq}^0(A^{2+}) = 200$ units
 $\lambda_{eq}^0(B^{3-}) = 400$ units
 $\lambda_{eq}^0(A_3B_2) = ?$

- (1) 600 units
- (2) 400 units
- (3) 200 units
- (4) 100 units

36) Consider the change in oxidation state of Bromine corresponding to different EMF values as



shown in the diagram below:
disproportion is :

Then the species undergoing

- (1) BrO_3^-
- (2) BrO_4^-
- (3) Br_2
- (4) HBrO

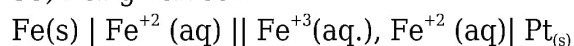
37)

Match the column :

	Column-I		Column-II
(A)	Leclanche cell (Dry cell)	(P)	Converts energy of combustion into electrical energy
(B)	Lead-storage battery	(Q)	Reaction at cathode : $\text{O}_2(\text{g}) + 4\text{H}^+(\text{aq}) + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}(\text{l})$
(C)	Fuel cell	(R)	Reaction at cathode : $\text{MnO}_2 + \text{NH}_4^+ + \text{e}^- \rightarrow \text{MnO}(\text{OH}) + \text{NH}_3$
(D)	Rusting	(S)	Reaction at anode : $\text{Pb}(\text{s}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{PbSO}_4(\text{s}) + 2\text{e}^-$

- (1) A-R, B-S, C-P, D-Q
- (2) A-S, B-R, C-P, D-Q
- (3) A-P, B-Q, C-R, D-S
- (4) A-P, B-R, C-Q, D-S

38) For given cell



E°_{cell} is 1.211 V, then correct value of ΔG° is -

- (1) -116.86 kJ
- (2) -233.7 kJ
- (3) 116.86 kJ
- (4) 233.7 kJ

39) Resistance of a conductivity cell (cell constant 129 cm^{-1}) filled with KCl solution of 0.1 M concentration is 100Ω , when same cell is filled with 0.2 M KCl solution resistance is 50Ω , then ratio of molar conductivity of 0.1 M and 0.2 M KCl solution is :

- (1) 2 : 1
- (2) 1 : 2
- (3) 1 : 1

(4) 4 : 1

40) Al_2O_3 is reduced by electrolysis at low potential and high current. If 4.0×10^4 A of current is passed through molten Al_2O_3 for 6 hours, what mass of aluminium is produced ? (Assume 100% current efficiency, At. mass of Al = 27 g mol^{-1})

- (1) $1.3 \times 10^4 \text{ g}$
- (2) $9.0 \times 10^3 \text{ g}$
- (3) $8.1 \times 10^4 \text{ g}$
- (4) $2.4 \times 10^5 \text{ g}$

41) Consider the following relations for emf of a electrochemical cell :

- (a) emf of cell = (Oxidation potential of anode) - (Reduction potential of cathode)
- (b) emf of cell = (Oxidation potential of anode) + (Reduction potential of cathode)
- (c) emf of cell = (Reduction potential of anode) + (Reduction potential of cathode)
- (d) emf of cell = (Oxidation potential of anode) - (Oxidation potential of cathode)

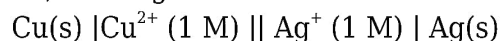
Which of the above relations are correct :

- (1) (a) and (b)
- (2) (c) and (d)
- (3) (b) and (d)
- (4) (c) and (a)

42) Limiting molar conductance of H^+ and CH_3COO^- ions are $344 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$ and $40 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$ respectively. Molar conductance of 0.008 M CH_3COOH is $48 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$. What will be the value of K_a for CH_3COOH :-

- (1) 1.4×10^{-5}
- (2) 1.2×10^{-5}
- (3) 1.4×10^{-4}
- (4) 1×10^{-5}

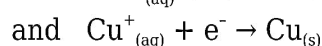
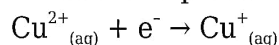
43) In the galvanic cell



the electrons will travel in the external circuit :

- (1) from Ag to Cu
- (2) from Cu to Ag
- (3) electrons do not travel in the external circuit
- (4) in any direction

44) The electrode potentials for

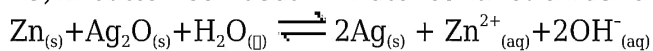


are +0.15 V and +0.50 V respectively. The value of $E^{\circ}_{\text{Cu}^{2+}/\text{Cu}}$ will be :-

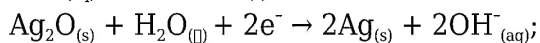
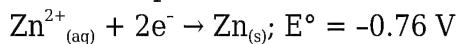
- (1) 0.500 V

- (2) 0.325 V
- (3) 0.650 V
- (4) 0.150 V

45) A button cell used in watches function as following



If half cell potentials are



$$E^{\circ} = 0.34 \text{ V}$$

then the cell potential will be :-

- (1) 1.34 V
- (2) 1.10 V
- (3) 0.42 V
- (4) 0.84 V

BIOLOGY

1) Life cycle of *Drosophila* is of :-

- (1) 2 years
- (2) 2 months
- (3) 2 weeks
- (4) 2 days

2) In F_2 -generation, genotypic and phenotypic ratios are same in case of :-

- (1) Co-dominance
- (2) Incomplete dominance
- (3) Test cross
- (4) All the above

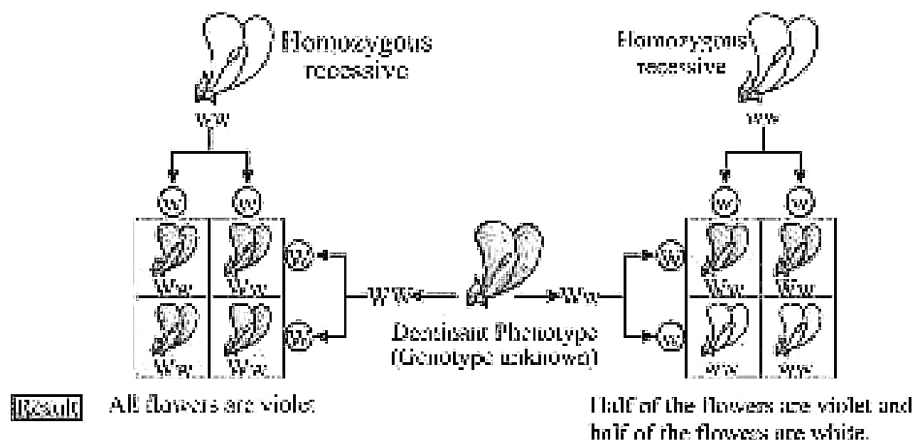
3) If Tt pea plant is crossed with Tt pea plant, then the offsprings will be -

- A- 75% plants are Tall
- B- 25% plants are Dwarf
- C- 50% plants are homozygous
- D- 50% plants are heterozygous

Select the correct option -

- (1) Only A, C, D
- (2) A, B, C, D
- (3) Only B, C, D
- (4) Only A, B, C

4) Observe the given diagram & select the correct option :-



- (1) It is the diagrammatic representation of a out cross.
- (2) It is the comparison of two monohybrid crosses, used to find out the phenotype of organisms.
- (3) It is the diagrammatic representation of reciprocal cross to find out the genotype of organism.
- (4) It is the diagrammatic representation of test cross to find out the genotype of dominant organisms.

5) With respect to ABO blood groups, how many genotypes and phenotypes, respectively, are possible in the population of human being?

- (1) 6 and 4
- (2) 4 and 6
- (3) 3 and 4
- (4) 4 and 8

6) Term Linkage was given by :-

- (1) Mendel
- (2) Morgan
- (3) Bateson
- (4) Boveri

7) How many types of gametes are formed from AaBbCc?

- (1) 4
- (2) 8
- (3) 16
- (4) 12

8) Dominance is not an autonomous feature of a gene or the product, when :-

- (1) More than one gene influence the same phenotype.
- (2) More than one phenotype is influenced by the same gene.
- (3) More than one genotype is influenced by the same gene.
- (4) More than one allele are there for a character.

9) A test cross enables us to -

- (1) determine the viability of cross
- (2) distinguish between homozygous dominant and heterozygous dominant
- (3) determine whether two species can interbreed
- (4) determine the similarities in the DNA of two species

10) Match the following (Column-I with Column-II)

Column-I		Column-II	
a	Allele	i	In an individual two alleles of a gene are identical
b	Genotype	ii	An individual having two different alleles of a gene
c	Homozygous	iii	Genetic constitution of an organism
d	Heterozygous	iv	Alternative forms of a gene

- (1) a-iv, b-ii, c-iii, d-i
- (2) a-iv, b-iii, c-i, d-ii
- (3) a-iii, b-iv, c-ii, d-i
- (4) a-iii, b-ii, c-iv, d-i

11) **Assertion (A):** Mendel's law of Independent assortment does not hold good for the genes that are located closely on the same chromosome.

Reason (R): Closely located genes always assort independently.

- (1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**
- (2) **(A)** is correct but **(R)** is not correct
- (3) **(A)** is not correct but **(R)** is correct
- (4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

12) Which combination of sex chromosome results in a female Grasshopper?

- (1) XX
- (2) XY
- (3) ZW
- (4) XO

13) Multiple alleles can be found during study of-

- (1) Gametes
- (2) Individual
- (3) Population
- (4) All of these

14) Recessive trait is seen due to :-

- (1) Normal Enzyme
- (2) Non-functional Enzyme
- (3) No Enzyme
- (4) Both (2) and (3)

15) Linkage can be broken by :-

- (1) Gene mutation
- (2) Segregation
- (3) Crossing-over
- (4) Independent assortment

16) Which of the following scientist explained Mendelian principles on the basis of chromosomal movement during meiosis?

- (1) T.H. Morgan
- (2) Sutton and Boveri
- (3) Alfred sturtevant
- (4) Jacob and Monad

17) Which of the following structure was discovered by Henking?

- (1) y-body
- (2) Bar body
- (3) x-body
- (4) Nu-body

18) Law of independent assortment was proposed on the basis of :-

- (1) Out cross
- (2) Reciprocal cross
- (3) Test cross
- (4) Dihybrid cross

19) Phenylketonuria is an example of -

- (1) Pleiotropism
- (2) Co-dominance
- (3) Linkage
- (4) All of these

20) In *Drosophila* the genes for eye colour and body colour are present on _____ and distance between the genes is _____ cM :-

- (1) X chromosome, 1.3
- (2) X chromosome, 37.2

- (3) Autosome, 1.3
- (4) Autosome, 37.2

21) Mark the **incorrect pair** with respect to pea plant :-

- (1) Round Seed – Dominant
- (2) Yellow Seed – Recessive
- (3) Violet Flower – Dominant
- (4) Yellow pod – Recessive

22) Which of the following cross give 9 : 3 : 3 : 1 phenotypic ratio ?

- (1) AABb × AaBB
- (2) AaBBCcDDEE × AaBBCcDDEE
- (3) AaBB × AaBb
- (4) aaBBCc × AAbbCc

23) What will be the progenies when there is marriage between colourblind father's son and colourblind mother's daughter?

- (1) 50% sons colour blind
- (2) 50% daughters carriers
- (3) 50% daughters normal
- (4) All of these

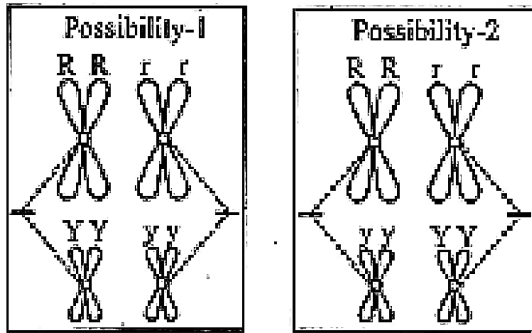
24) A disease in which a protein that is the part of red and green colours perception is affected. This disease is :-

- (1) y- linked recessive disorder
- (2) x - linked recessive disorder
- (3) x - linked dominant disorder
- (4) Autosomal recessive disorder

25) *Drosophila melanogaster* is related to-

- (1) Linkage
- (2) Chromosomal map
- (3) Discovery of x - body
- (4) Both (1) and (2)

26) Given diagram shows :-



- (1) Dominance
- (2) Mitosis
- (3) Independent assortment
- (4) Linkage

27) Mendel's work remained unrecognised for long time due to :-

I. Communication was not easy.

II. Concept of factors which did not blend was not accepted

III. Use of mathematics to explain biological problem was unacceptable

IV. He could not provide any physical proof for the existence of factors.

Choose the right combinations

- (1) I and II
- (2) II and III
- (3) III and IV
- (4) All of these.

28) If size of starch grain in pea is considered as phenotype, the heterozygous condition of Alleles (Bb) will show.

- (1) Incomplete dominance
- (2) Co-dominance
- (3) Dominant phenotype
- (4) Recessive phenotype

29) Select the mis - matched pair

- (1) Klinefelters syndrome - Gynaecomastia
- (2) Turner's syndrome - Rudimentary ovaries
- (3) Down's syndrome - observed only in male
- (4) Haemophilia - Extremely rare in females.

30) Inheritance of skin colour in human beings is an example of :-

- (1) Pleiotropism gene
- (2) Monogenic inheritance
- (3) Polygenic inheritance

(4) Mendelian inheritance

31) Who observed that behaviour of chromosomes at meiosis can serve as the cellular basis of both segregation and independent assortment ?

- (1) Sutton & Boveri
- (2) Banden and Boveri
- (3) W. Fleming
- (4) Boveri and Braurer

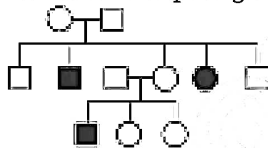
32) Which among the following show polygenic inheritance?

- (1) Starch grain size in pea
- (2) Height in humans
- (3) Flower colour in pea
- (4) ABO Blood group in humans

33) According to a cross performed by Morgan in *Drosophila*, what was the percent recombination between the genes for white eye and miniature wings ?

- (1) 37.2%
- (2) 98.7%
- (3) 1.3%
- (4) 68.2%

34) Given below is the pedigree of sickle cell anaemia, in a family



In this the RBC of both parents will be -

- (1) Normal
- (2) Sickle shaped
- (3) Both normal & sickle shaped
- (4) Cannot be determined

35) Which is **incorrectly matched** ?

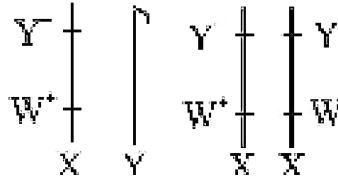
- (1)  - Unaffected male
- (2)  - Second marriage
- (3)  - Affected female
- (4)  - Sex unspecified

36)

The haploid- diploid sex determination system, as seen in honey bee has characteristic features such as :-

- (1) Males produce sperm by mitosis
- (2) Worker bee are developed from unfertilized eggs
- (3) Males have fathers and thus can have sons.
- (4) Males do not have grandfather and so can not have grandsons.

37) How many types of gametes will be produced by male and female *Drosophila* respectively having



following arrangement of two genes ?

- (1) Two & Two
- (2) Two & Four
- (3) One & Two
- (4) Four & Four

38) Match the following column-I with column-II :-

Column-I		Column-II	
A	Test cross	1	9 : 3 : 3 : 1
B	Monohybrid cross	2	Tt × tt
C	Out cross	3	Tt × TT
D	Dihybrid cross	4	3 : 1

Codes -

- | | | | | |
|-----|---|---|---|---|
| | A | B | C | D |
| (1) | 2 | 4 | 3 | 1 |
| (2) | 2 | 4 | 1 | 3 |
| (3) | 3 | 4 | 1 | 2 |
| (4) | 1 | 4 | 2 | 3 |
- (1) 1
 - (2) 2
 - (3) 3
 - (4) 4

39) Read the following statements carefully :

- (i) Genes located on same locus but having different expressions are called polygenes.
- (ii) A cross between a hybrid and its recessive parent is called test cross.
- (iii) Mendel proposed that something was being stably passed down unchanged from parents to offsprings called factors.
- (iv) Four types of gametes are produced by individuals of AABbcc genotype.

How many statements are **correct** from these ?

- (1) Four
- (2) Three
- (3) Two
- (4) One

40) **Assertion :-** Punnett square is a graphical representation to calculate the probability of all possible genotypes of offsprings in a genetic cross.

Reason :- It was developed by a British geneticist, Reginald C. Punnett.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

41) **Assertion :-** When both the genes in a dihybrid cross are situated on the same chromosome, usually the proportion of parental gene combinations are much higher than the non-parental type.

Reason :- Genes present on same chromosome may show the physical association or linkage.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

42) How many true breeding pea plant varieties were selected by Mendel for his experiments, as pairs which were similar except for one character with contrasting trait?

- (1) Seven
- (2) Twelve
- (3) Fourteen
- (4) Twenty

43) Mark the correct phenotype ratio obtained for plants with round yellow and wrinkled green seeds in F_2 generation in the dihybrid cross that was conducted by Mendel.

- (1) 3 : 1
- (2) 9 : 1
- (3) 1 : 3
- (4) 1 : 1

44) **Statement-I:-** The inheritance of flower colour in dog flower is a good example to understand incomplete dominance

Statement-II:- ABO blood group is good example of multiple allele.

- (1) Both **Statement I** and **Statement II** are incorrect.
- (2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct

(4) Both **Statement I** and **Statement II** are correct.

45) Read the following statements and find out **incorrect** one/ones.

(I) Flower colour in *Antirrhinum* species shows codominance.

(II) ABO blood group may show complete dominance.

(III) A single gene product may produce more than one effect.

(IV) Wrinkled seed shape in garden pea is dominated by round seed shape.

(1) Only I

(2) Only II

(3) Only III

(4) Both I and IV

46) **Assertion :-** Using conventional methods of diagnosis early detection is not possible.

Reason :- Recombinant DNA technology, PCR & ELISA are some of technique that serve the purpose of early diagnosis.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

47) **Assertion :-** Toxicity testing in transgenic animals allow us to obtain results in less time.

Reason :- Transgenic animals are made that carry genes which make them more sensitive to the toxic substances than non-transgenic animals.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

48) **Assertion :-** In gene therapy of ADA, a patient requires periodic infusion of genetically engineered lymphocytes.

Reason :- If the correct ADA gene is introduced into cells at early embryonic stage, it could be a permanent cure.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

49) Match the following :-

	Column-I		Column-II
--	-----------------	--	------------------

(A)	Stanley Cohen and Herbert Boyer	(i)	Separation of DNA fragments
(B)	Hind II	(ii)	Construction of first recombinant DNA
(C)	Gel electrophoresis	(iii)	First restriction endonuclease
(D)	Bam HI site	(iv)	At tetracycline resistance gene in vector pBR322

- (1) A-II, B-III, C-IV, D-I
- (2) A-II, B-III, C-I, D-IV
- (3) A-II, B-IV, C-I, D-III
- (4) A-IV, B-III, C-II, D-I

50) Blunt end producing restriction enzyme is :

- (1) EcoRI
- (2) EcoRV
- (3) Bam HI
- (4) Hind III

51) Which technique is employed to check the progression of a restriction enzyme digestion ?

- (1) Polymerase Chain Reaction
- (2) Restriction mapping
- (3) Downstream processing
- (4) Agarose gel electrophoresis

52) Palaeontologists unearthed a human skull during excavation. A small fragment of the scalp tissue was still attached to it. Only little DNA could be extracted from it. If the genes of ancient man need to be analysed the best way of getting sufficient amount of DNA from this extract is-

- (1) Subjecting the DNA to gel electrophoresis
- (2) Treating the DNA with restriction endonuclease
- (3) Hybridising the DNA with a DNA probe
- (4) Subjecting the DNA to PCR

53)

Read the following statements carefully and select the correct option :-

- (A) Source of the restriction enzyme Hind III is *Haemophilus influenzae*.
- (B) In biolistic method of gene transfer, microparticles made up of gold or tungsten are coated with foreign DNA.
- (C) Micro-injection method for injecting recombinant DNA is used for animal cell.
- (D) Primers are chemically synthesized oligonucleotides that are complementary to the regions of

DNA in PCR.

How many of the above statements are correct?

- (1) Four
- (2) Three
- (3) Two
- (4) One

54) A transgenic food crop which may help in solving the problem of acquired night blindness in developing countries is :-

- (1) *Bt*-rice
- (2) *Bt*-soyabean
- (3) Golden rice
- (4) *Flavr savr* tomatoes

55) Treatment of a genetic disorder by manipulating genes is called ?

- (1) rDNA technology
- (2) Enzyme replacement therapy
- (3) Gene therapy
- (4) Bone marrow transplantation

56) The Indian parliament has recently cleared which amendment of the Indian patents bill :-

- (1) 1st
- (2) 2nd
- (3) 3rd
- (4) 4th

57) Which step proved to be the main challenging obstacle in the production of human insulin by genetic engineering?

- (1) Splitting A & B polypeptide chains
- (2) Addition of C-peptides to pro-insulin
- (3) Getting insulin assembled into mature form
- (4) Removal of D-peptide from immature insulin

58) Given below are two statements : One is labelled as Assertion (A) and the other is labelled as Reason (R) :

Assertion (A) : Bt toxin kills certain insect such as lepidopterans, coleopterans and dipterans but not human being.

Reason (R) : Bt toxin protein exist as inactive protoxins, it is converted into an active form due to alkaline pH of the gut of insect.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both (A) and (R) are correct and (R) is the correct explanation of (A)
- (2) Both (A) and (R) are correct and (R) is not the correct explanation of (A)
- (3) (A) is correct but (R) is not correct
- (4) (A) is not correct but (R) is correct

59) **Assertion** : RNAi take place in all eukaryotic organism as a method of cellular defense.

Reason : This method involves silencing of a specific mRNA due to complementary ds RNA molecule that binds to and prevents translation of the mRNA (silencing).

- (1) Both Assertion and Reason are True and the Reason is a correct explanation of the Assertion.
- (2) Both Assertion and Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion and Reason are False.

60) Single stranded DNA or RNA segment tagged with a radioactive molecule is called :

- (1) Clone
- (2) Plasmid
- (3) Vector
- (4) Probe

61) First recombinant DNA was constructed in the year :-

- (1) 1963
- (2) 1974
- (3) 1981
- (4) 1972

62) **Assertion** :- In order to force bacteria to take up the plasmid, the bacteria cells must first be made 'competent' to take up DNA.

Reason :- DNA is a hydrophilic molecule, it cannot pass through cell membranes.

- (1) Both Assertion and Reason are True and the Reason is a correct explanation of the Assertion.
- (2) Both Assertion and Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion and Reason are False.

63) A pathogen of several dicot plants which is able to deliver a piece of DNA known as 'T-DNA' to transform normal plant cells into a tumour, is :-

- (1) Retrovirus
- (2) Baculovirus
- (3) *Bacillus thuringiensis*
- (4) *Agrobacterium tumifaciens*

64) In restriction enzymes (like EcoRI) the roman number indicate the :-

- (1) Order in which the enzyme were discovered from the strain of bacteria
- (2) Order in which the enzymes were isolated from the strain of bacteria.
- (3) Genus of the prokaryotic cell or bacteria
- (4) Strain of the bacteria

65) Which enzyme is not related correctly with its function ?

	Column A	Column B
(1)	<i>Taq</i> polymerase	Synthesis of DNA
(2)	Chitinase	Obtained DNA by lysis of bacterial cell
(3)	Exonuclease	Remove nucleotides from the ends of DNA
(4)	DNA ligase	Joining the fragments of DNA

- (1) 1
- (2) 2
- (3) 3
- (4) 4

66) Which of the following is the correct match ?

- (1) *Cry IAb* - Cotton bollworms
- (2) *Cry IAc* - Cotton bollworms
- (3) *Cry IAc* - Corn borer
- (4) *Cry IIAb* - Corn borer

67) Which enzyme is responsible for restricting the growth of bacteriophage in *E.coli* ?

- (1) Restriction endonuclease
- (2) Gyrase
- (3) DNA ligase
- (4) DNA polymerase

68) Which of the following step in PCR (polymerase chain reaction) is enzyme catalysed ?

- (1) Denaturation of DNA
- (2) Annealing of primers to template DNA strand
- (3) Extension of primer end on the template DNA strand
- (4) All the above

69) **Statement-1** : Restriction enzymes cut the strand of DNA a little away from the centre of the palindromic site but between same 2 bases on the opposite strands.

Statement-2 : Each restriction endonuclease function by inspecting the length of DNA sequence.

- (1) Both statements are incorrect
- (2) Statement-1 correct but statement-2 incorrect

(3) Both statements are correct

(4) Statement-2 correct but statement-1 incorrect

70) Match the column -

Column-I		Column-II	
(a)	Gel electrophoresis	(i)	Bright orange coloured
(b)	PCR	(ii)	Chilled ethanol
(c)	DNA precipitation	(iii)	Sieving effect
(d)	Ethidium bromide stained DNA	(iv)	DNA primer

(1) a - i; b - ii, c - iii, d - iv

(2) a - ii; b - iii, c - iv, d - i

(3) a - iii; b - iv, c - i, d - ii

(4) a - iii; b - iv, c - ii, d - i

71) After the formation of the product in the bioreactors, it undergoes through separation and purification processes before a finished product is ready for marketing. These processes are collectively referred to as :

(1) Upstream processing

(2) Downstream processing

(3) Elution

(4) Transformation

72) **Assertion :** Normally unless one cuts the vector and the source DNA with the same restriction enzyme, the recombinant DNA molecule cannot be created.

Reason : Cutting of the vector and source DNA by the same restriction enzyme create sticky ends that facilitates the action of enzyme DNA ligase.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

73) Which of the following vector is used extensively for genetic engineering in animals?

(1) *E.coli*

(2) *Ti* plasmid

(3) *Bacillus thuringiensis*

(4) Retrovirus (Disarmed)

74) Which of the following statement are incorrect ?

- (a) RNAi takes place in all eukaryotic organisms as a method of cellular defence.
- (b) C peptide along with A and B peptides are present in mature insulin.
- (c) In 1990, the first transgenic cow, Rosie, produced human protein-enriched milk.
- (d) Recombinant DNA technology serve the purpose of early diagnosis.
- (e) 27 Documented Basmati rice varieties are grown in India.

- (1) b, c, d
- (2) a, b, c
- (3) a, d, e
- (4) b and c only

75) Which novel strategy was adopted against nematode *Meloidogyne incognitia* which infect the roots of tobacco plants?

- (1) Insertional inactivation
- (2) RNA interference (RNAi)
- (3) Gene therapy
- (4) Auto radiography

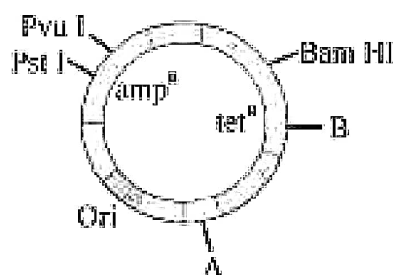
76) **Assertion** :- PCR is now routinely used to detect HIV in suspected AIDS patients.

Reason :- PCR is being used to detect mutation in genes in suspected cancer patients.

- (1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Both Assertion & Reason are False.

77) How many DNA molecules will be produced from one DNA molecule after 6 PCR cycles?

- (1) 10
- (2) 15
- (3) 32
- (4) 64



78) In the following diagram find out A and B :

- (1) A $\Rightarrow$ rop, B = Sal I
- (2) A $\Rightarrow$ Sal I, B = rop
- (3) A $\Rightarrow$ EcoRI, B $\Rightarrow$ rop
- (4) A $\Rightarrow$ rop, B = Cla I

79) Which of the following statements are correct ?

- (a) *Agrobacterium*-mediated gene transfer involves the use of a bacterium to transfer genes into dicot plant cells.
- (b) DNA polymerase derived from *Thermus aquaticus* is used in PCR due to its heat stability.
- (c) Bioreactors are used for large scale cultivation of micro-organism for biotechnological processes.
- (d) 2,00,000 varieties of basmati rice are grown in India.
- (e) The sieving effect during gel electrophoresis is due to ethidium bromide.

- (1) a, b, e
- (2) b, c, d
- (3) a, b, c
- (4) d and e only

80) _____ is a procedure through which a piece of DNA is introduced in a host bacterium.

- (1) Transformation
- (2) Cloning
- (3) Polymerase chain reaction
- (4) Downstream processing

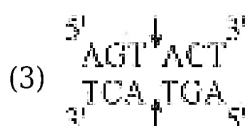
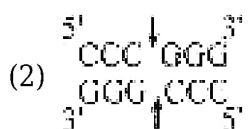
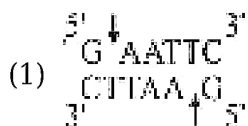
81) During the isolation of DNA from a bacterial cell, which of the following enzyme is not used?

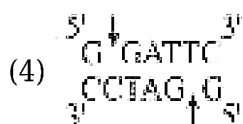
- (1) Ribonuclease
- (2) Deoxyribonuclease
- (3) Protease
- (4) Lysozyme

82) First clinical gene therapy was given in :

- (1) 1997
- (2) 1990
- (3) 1977
- (4) 1980

83) Which of the following palindromic sequence is recognised by EcoRI ?





84) Recombinant DNA molecule can be created normally by cutting the vector DNA and source DNA respectively with :

- (1) Hind III, Hind III
- (2) Hind II, Alu I
- (3) Hind II, EcoR I
- (4) Hind II, BamH I

85) Fill up the blanks :

- A. The construction of the first recombinant DNA emerged from the possibility of linking a gene encoding antibiotic resistance with a native..... of.....
- B. Hind II always cut DNA molecules at a particular point by recognising a specific sequence ofbase pairs. This specific sequence is known as..... for Hind II.
- C. The.....in DNA is a sequence of base that reads same on the two strands when orientation of reading is kept the same.
- D., a pathogen of several dicot plants is able to deliver a piece of DNA known as..... to transform normal plant cells into tumor cells.

- (1) A - Plasmid, *Salmonella typhimurium*; B - Six, Recognition sequence, C - Palindrome, D - *Agrobacterium tumifaciens*, T-DNA
- (2) A - Plasmid, *Salmonella typhimurium*; B - Seven, Recognition sequence, C - Palindrome, D - *Agrobacterium tumifaciens*, T-DNA
- (3) A - Chromosome, *Salmonella typhimurium*; B -Six, Recognition sequence, C - Palindrome, D - *Agrobacterium tumifaciens*, T-DNA
- (4) A - Plasmid, *Salmonella typhimurium*; B - Six, Recognition sequence, C - Palindrome, D - *Agrobacterium tumifaciens*, Z-DNA

86)

Given below are two statements :-

Statement-I :- Restriction endonucleases do not cut the DNA of own cell in Prokaryotes.

Statement-II :- Prokaryotic DNA does not have sites of restriction endonuclease.

Choose the correct answer from the options give.

- (1) Both statement I and statement II are correct.
- (2) Only statement I is correct.
- (3) Only statement II is correct
- (4) Both statement I and statement II are incorrect.

87) **Statement I** :- The normal *E.coli* cells do not carry resistance against ampicillin, chloramphenicol, tetracycline or kanamycin etc.

Statement II :- The presence of a chromogenic substrate gives blue coloured colonies if the plasmid in the bacteria does not have an insert.

- (1) Statement I and II both are correct.
- (2) Statement I and II both are incorrect.
- (3) Only Statement I is correct.
- (4) Only Statement II is correct.

88) The use of bioresources by multinational companies and other organization without proper authorisation from countries and people concerned without compensatory payment, it is known as :-

- (1) Biopiracy
- (2) Biopatent
- (3) Biowar
- (4) Bioremediation

89) **Statement-I :-** At present 30 recombinant therapeutic have been marketed for human use in India.

Statement-II :- Recombinant therapeutic do not induce unwanted immunological response as is common in case of similar product isolated from non human source
Choose the correct answer from option given below :

- (1) Statement I and II both are correct
- (2) Statement I and II both are incorrect
- (3) Statement-I is correct & Statement-II is incorrect.
- (4) Statement-I is incorrect & Statement-II is correct.

90) **Statement-I :** Transgenic mice are being used to test the safety of the polio vaccine if successful and found to be reliable they could replace the use of monkeys.

Statement-II : 95% of all existing Transgenic animals are mice.

- (1) Both Statement-I and Statement-II are incorrect.
- (2) Statement-I is correct but Statement-II is incorrect.
- (3) Statement-I is incorrect but Statement-II is correct.
- (4) Both Statement-I and Statement-II are correct.